

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 10%

QUESTIONS		
Q. No.	Question	Bloom's Level
1	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand
2	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand
3	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand
4	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember
5	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand

Code of Conduct:

/ VENNA MEENAKSHI REDDY - 192471048 (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: VENNA MEENAKSHI REDDY

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TECHNICAL PRESENTATION

CSA0509 – Database Management Systems

Presented by:

VENNA MEENAKSHI REDDY (192471048)



1. Create and present an ER diagram for an Online Library System, explaining each component clearly.



Introduction



Online Library System : An Online Library System is a database application used to manage books, members, borrowing, and returns.

Main Components

- Entities
- Attributes
- Primary Keys
- Relationships

Purpose

- Organizes library data efficiently.
- Tracks books and members.



Explanation of Components



Entities

- **Member** – Stores member information.
- **Book** – Stores book details.
- **Loan** – Stores issue and return details.

Relationships

- Member **borrow**s Book.
- Loan records each borrowing transaction.

Attributes

Member

Member_ID (PK)

Name

Email

Phone

Book

Book_ID (PK)

Title

Author

Category

Loan

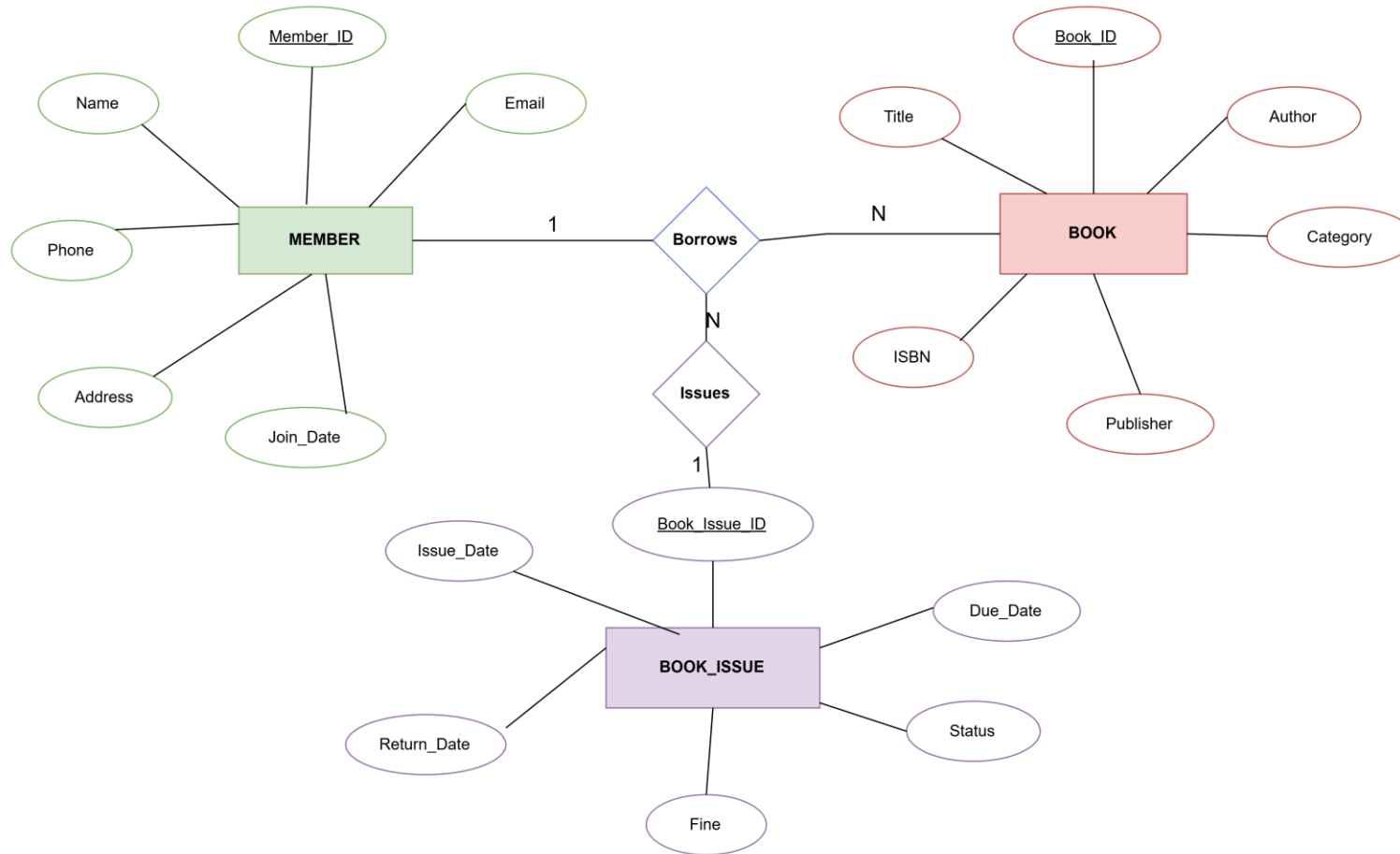
Loan_ID (PK)

Issue_Date

Return_Date

Fine

ER Diagram



- ❑ **Member**, **Book**, and **Book_Issue** are the main entities.
- ❑ **Member** stores member details, and **Book** stores book information.
- ❑ **Book_Issue** records issue date, return date, due date, and fine.
- ❑ **Borrows** and **Issues** define the relationships between the entities.
- ❑ **Member_ID**, **Book_ID**, and **Book_Issue_ID** are the **Primary Keys** that uniquely identify each record.

Fig: ER Diagram of Online Library System



Conclusion



- The ER diagram provides a clear structure for the Online Library System.
- It uses entities, attributes, and relationships to organize library data efficiently.
- It helps reduce data redundancy and improve data accuracy.
- It simplifies book management, member management, and loan tracking.
- It supports an organized, scalable, and efficient database system.



2. Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.



Introduction



Definition

The **Enhanced Entity Relationship (EER) Model** is an extension of the ER model that represents complex real-world data more effectively. It introduces **generalization**, **specialization**, and inheritance to model entities with common and unique characteristics.

Why EER is Needed

- Represents complex organizational structures.
- Eliminates duplicate data.
- Improves database design.
- Supports inheritance of common attributes.
- Makes the database easier to maintain and expand.



Generalization



Definition

Generalization is the process of combining multiple lower-level entities into a higher-level entity by identifying their common attributes.

Example

Student + Faculty + Staff $\rightarrow$ Person

Advantages

- Reduces redundancy
- Improves reusability
- Simplifies database design



Specialization



Definition

Specialization is the process of dividing one higher-level entity into multiple lower-level entities based on distinguishing characteristics.

Example

Person → Student, Faculty, Staff

Advantages

- Stores specific information efficiently
- Improves accuracy
- Supports different operations for each entity



Specialization and Generalization in EER Model

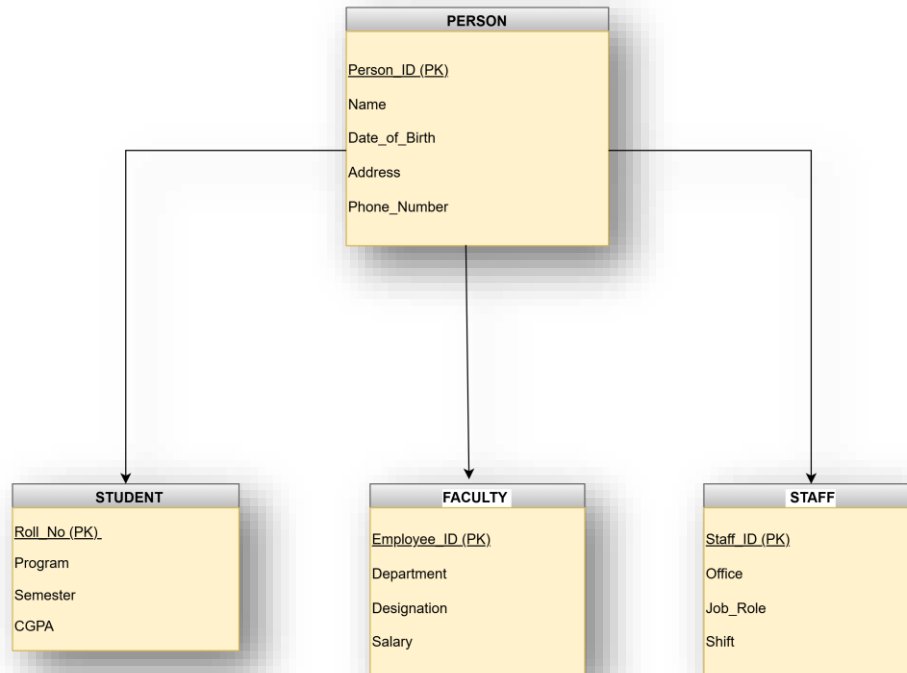
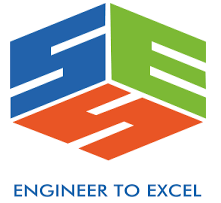


Fig: EER Model Showing Specialization

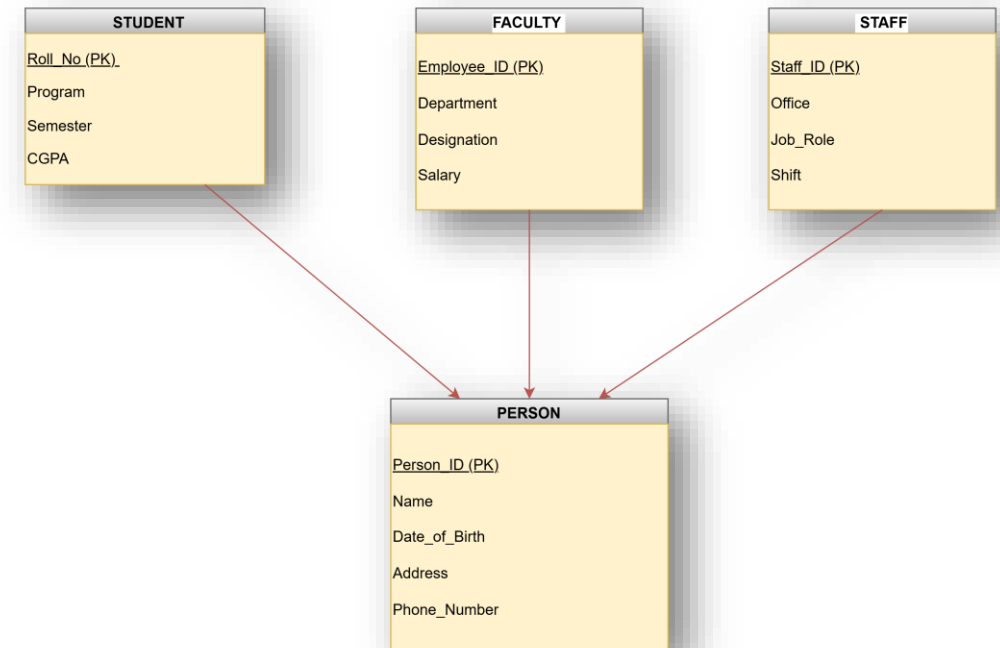


Fig: EER Model Showing Generalization



Conclusion



- The Enhanced Entity Relationship (EER) Model extends the traditional ER model by introducing generalization and specialization, enabling efficient representation of complex real-world data.
- In the university case study, the Person superclass and Student, Faculty, and Staff subclasses reduce redundancy, improve data organization, and simplify database maintenance.
- Therefore, the EER model is a powerful approach for designing scalable and efficient database systems.



3. Present the concept of data independence (logical and physical) and its significance in DBMS architecture.



Introduction

Definition

Data Independence is the ability to modify the database schema without affecting application programs or users.

Types

- Logical Data Independence
- Physical Data Independence

Why it is Important

- Simplifies database maintenance
- Reduces application changes

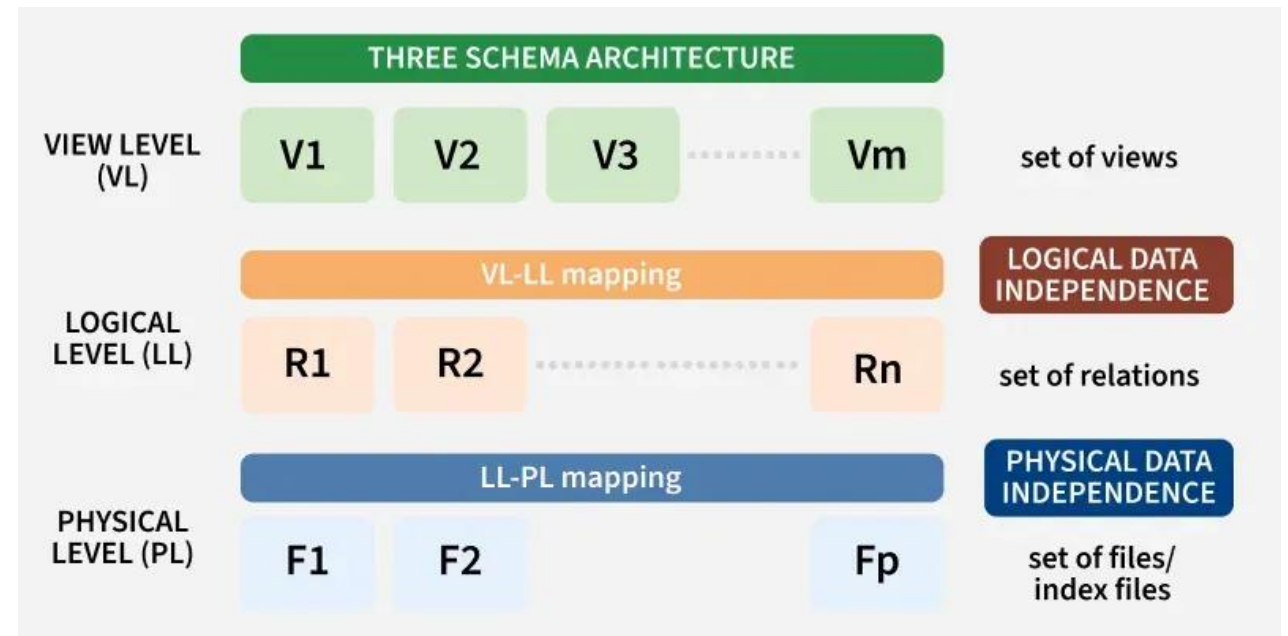


Fig: Logical and Physical Data Independence



Logical Data Independence

Definition

- The ability to change the **logical (conceptual) schema** without affecting user applications.

Changes Allowed

- Add or remove attributes
- Add new tables
- Modify relationships

Real-World Example

- A university adds an **Email** column to the **Student** table. Existing applications continue to work without modification.



Physical Data Independence

Definition

- The ability to change the **physical storage** of the database without affecting the logical schema or applications.

Changes Allowed

- Change file organization
- Create indexes
- Move data to new storage devices

Real-World Example

- A university database is moved from an HDD to an SSD for faster performance. Users and applications experience no changes.



Significance in DBMS Architecture



- Reduces data redundancy
- Improves database performance
- Simplifies maintenance
- Supports scalability
- Protects application programs from database changes
- Increases system reliability



4. Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.



Introduction



Software Components of a DBMS

Definition

- A Database Management System (DBMS) consists of software components that store, process, and manage data efficiently.

Main Components

- Query Processor
- Storage Manager
- Transaction Manager

Real-World Example: Banking System



Query Processor

Definition

- The Query Processor interprets and executes SQL queries submitted by users.

Functions

- Parses SQL queries
- Optimizes query execution
- Executes queries
- Returns results to the user

Example

`SELECT * FROM Student WHERE Department = 'CSE';`

- The Query Processor finds the best way to retrieve the required student records.



Storage Manager



Definition

- Manages how data is stored, retrieved, and updated on disk.

Functions

- Stores database files
- Manages indexes
- Retrieves and updates data
- Allocates storage space

Example: Saves and retrieves student records from the database.



Transaction Manager



Definition

- Controls database transactions and maintains data consistency.

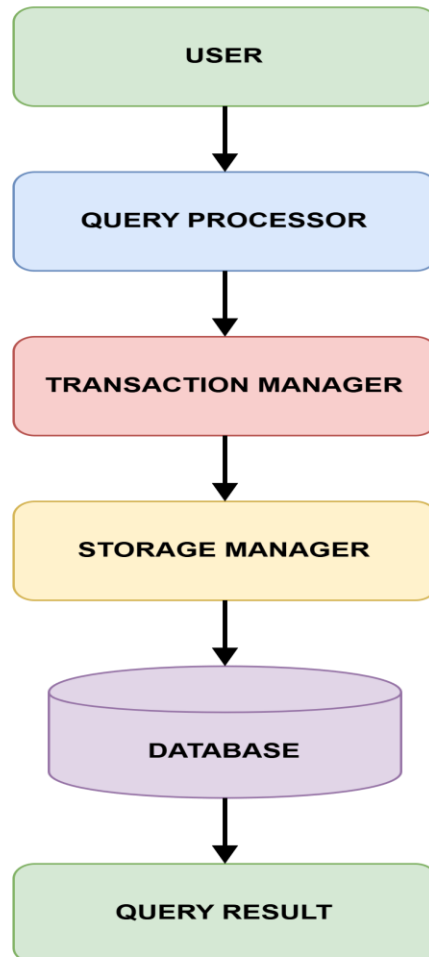
Functions

- Begins and ends transactions
- Supports Commit and Rollback
- Handles concurrent users
- Ensures ACID properties

Example: During online fee payment, if the payment fails, the transaction is rolled back to prevent incorrect data.

Working & Conclusion

WORKFLOW



CONCLUSION

- The Query Processor, Storage Manager, and Transaction Manager are the core software components of a DBMS.
- Together, they process queries, manage database storage, and ensure reliable transactions, making database systems efficient, secure, and scalable.



5. Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.



Introduction



Definition

- An **Entity** represents a real-world object in a database.
- **Types**
 - Strong Entity
 - Weak Entity

Importance

- Represents real-world data accurately.
- Defines relationships between entities.
- Improves database design and integrity.

Weak Entity

- A weak entity **does not have a complete primary key** and depends on a strong entity for identification.

Characteristics

- Depends on a Strong Entity.
- Uses a Partial Key.
- Represented by a double rectangle in an ER diagram.

Real-World Example

Dependent (Employee Database)

- Dependent_Name (Partial Key)
- Age
- Relationship

Depends on **Employee (Employee_ID)**.

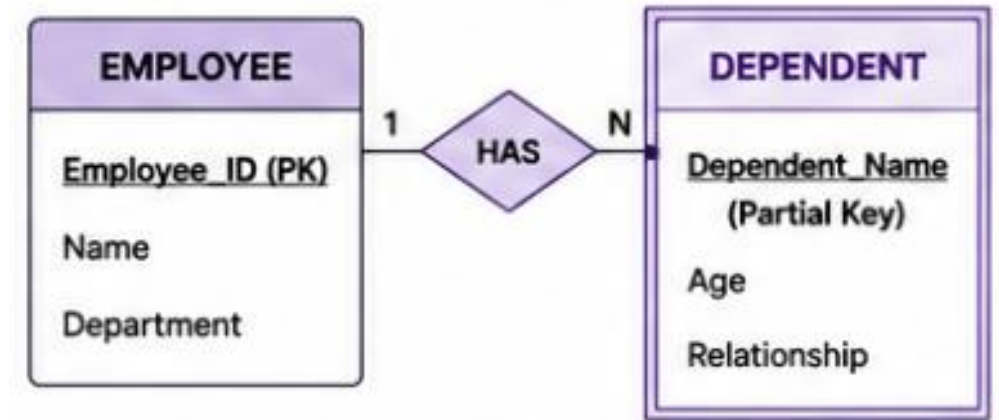


Fig: ER Diagram Showing Weak Entity



Strong Entity

Definition

- A strong entity has its own primary key and exists independently.

Characteristics

- Has a unique Primary Key.
- Does not depend on another entity.
- Represented by a single rectangle in an ER diagram.

Real-World Example

- Student
- Student_ID (PK)
- Name
- Department

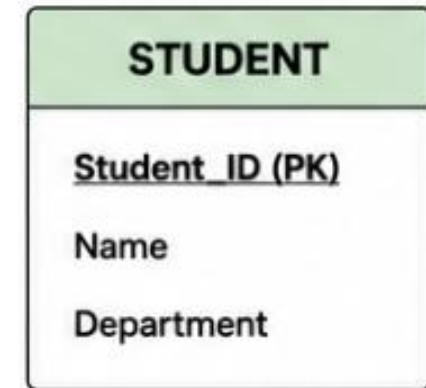


Fig: ER Representation of a Strong Entity



Conclusion



- Strong entities can exist independently with their own primary keys.
- Weak entities rely on strong entities for identification.
- Using both entities helps design accurate, organized, and efficient databases that represent real-world relationships.

THANK YOU